Investigating Circadian Advantages in the National Football League: A Case Study of West Coast Teams

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Abstract: Previous research in the National Football League (NFL) has shown that teams traveling eastward may gain circadian advantages during competition. The current study examines thirty years of performance using all West Coast teams subject to frequent travel in the NFL. We hypothesized that the West Coast teams would gain circadian advantages during eastward travel compared to travel within the same time zone, and when playing at their circadian peaks during the afternoon. Data for all away games played from the 1990 to 2019 seasons were collected from Pro-Football-Reference and FiveThirtyEight (n = 1,018). We investigated the impact of time of game, time zone, and their interaction on game outcomes, points scored and allowed, interceptions, sacks, punts, and completion, field-goal, and extra-point percentages. A series of generalized regressions with Bonferroni-corrected post-hoc tests were performed, controlling for day of the game and each team's strength using Elo rating. We found that teams experienced greater competitive advantages the further eastward they traveled, winning more games, and scoring the most points in the Eastern time zone. Our results also illustrated that teams scored more, allowed fewer points, and had higher completion percentages during afternoon games. Our study provides further evidence for the circadian rhythm advantage for West Coast teams. Limitations, implications, and future directions in the context of the NFL are also discussed.

Keywords: Sports Performance, Professional Football, Travel, Circadian Rhythm

Introduction and Background

All thirty-two teams in the National Football League (NFL) frequently travel across time zones in the United States, with games occurring approximately once a week for seventeen weeks. However, there is an exception of one bye week for each team, in which no game is scheduled. Previous research has shown that traveling across time zones results in a disruption to a team's circadian rhythm, a daily fluctuation in cognitive and physical performance (Jehue, Street, and Huizenga 1993). Circadian rhythms are 24-hour cycles that are part of the body's internal clock that specifically help regulate sleep cycles and other bodily functions (National Institute of General Medical Sciences 2021). During the day, circadian rhythms are sensitive to light exposure and send signals that promote alertness. During the night, in response to darkness, they signal for melatonin production to promote and maintain sleep (Sleep Foundation 2020). When the circadian rhythm is disrupted due to travel, it can lead to fatigue and alertness decrements (Sleep Foundation 2020), as may also be measured by the physical performance of athletes. Since 69 percent of NFL teams are coastally located (NFL 2021), the majority of the NFL must cope with the deleterious effects of travel.

Traveling across multiple time zones disrupts athletes' natural internal clocks, creating a circadian advantage for those playing at times closer to peak performance time. A circadian peak exists in the late afternoon for strength, flexibility, and reaction time (Smith, Guilleminault, and Efron 1997; Facer-Childs, Boiling, and Balanos 2018). Accordingly, there is a decrease in performance during evening periods (Smith, Guilleminault, and Efron 1997; Carrier and Monk

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2000). Likewise, due to the physically taxing nature of the game of football, strength, flexibility, and reaction time are extremely crucial. A few of their many roles in the sport include the following: (1) strength is essential for ball carriers to withstand tackling attempts and frequent collisions from opposing players; (2) flexibility is vital for quarterbacks who seek to evade defensive players attempting to tackle them while maintaining mobility and throwing the football; and (3) reaction time is critical to offensive linemen who seek to physically block the defensive linemen from reaching the quarterback immediately after the ball is snapped.

Previous research has illustrated that West Coast teams in the NFL may possess a circadian advantage in terms of overall winning percentage when traveling eastward, as teams play closer to the peak time, as opposed to East Coast teams traveling westward (Smith, Guilleminault, and Efron 1997). This phenomenon is not just limited to the NFL, as it is seemingly present in multiple professional sports. For instance, within the National Basketball Association (NBA), all thirty teams travel frequently across time zones, which becomes even more imperative to manage during the playoffs. Previous work by Pradhan, Chachad, and Alton (2022), which examined nearly 500 games during the 2013–2014 to 2018–2019 playoffs, has shown that NBA teams traveling eastward scored more points than those traveling within the same time zone. The importance of travel on sports performance has also been highlighted in other professional sports contexts, such as Major League Baseball (MLB), and some studies have even considered the influence of the chronotype of athletes in this relationship (i.e., individual preferences for various activities including sleep and wakefulness based on one's circadian rhythm; Martínez-Pérez et al. 2020). For instance, East Coast teams like the Baltimore Orioles may travel to the West Coast to play an opponent such as the Los Angeles Angels at a 2:00pm start time. Due to the difference in time zones, players on the Orioles may feel like the game occurs at 5:00pm. In connecting individual chronotype, some studies have reported that athletes who perceive themselves as 'night owls' have experienced a higher batting average (.267) compared to those who believed they were 'early birds', with a batting average of (.259) in such games that started during this late afternoon window (Dell'Amore 2011). This advantageous performance is consistent with the circadian peak performance time of the late afternoon (Smith, Guilleminault, and Efron 1997).

Furthermore, Roy and Forest (2018) examined data from all teams within the NBA, NFL, and National Hockey League (NHL) over a period of five years. Their results revealed that the direction of travel may affect winning percentage, with an advantage to teams traveling eastward and disadvantage to those traveling westward. However, winning percentage is not the only factor indicative of performance, and it can naturally vary throughout the season as a result of factors like personnel changes or injuries. In addition, inspecting teams from varying locations in each of the three leagues without controlling for team factors introduces variability in schemes, traveling patterns, and scheduling, which are dependent on location, and can impact performance.

Purpose and Context

Thus, the current manuscript serves as a case study of only West Coast teams over a period of thirty years, as these teams often travel the most during the seasons. For example, all five West Coast NFL Teams (i.e., the San Francisco 49ers, Los Angeles Rams, Seattle Seahawks, and Los Angeles Chargers) often rank close to the top for total distance traveled during the regular season (DaSilva 2020; Kleinschmidt 2021). In addition, previous research has not comprehensively considered the quality of opponent when performing similar analyses. Consequently, our study accounts for the strength of a team and their opponent by utilizing a metric termed "Elo," which assigns a power rating to teams and is adjusted prior to each weekly matchup. Elo accounts for the expectation of each game outcome based on the difference in quality between the two teams involved, the winning margin, changes at the starting quarterback position, location of the matchup (including travel distance), and any additional rest days either team had coming into the game (Fischer-Baum et al. 2021).

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As previously mentioned, many studies have simply examined the effect of circadian disruption on performance by inspecting changes in winning percentages and points scored as performance-based metrics (Jehue, Street, and Huizenga 1993; Smith, Guilleminault, and Efron 1997; Roy and Forest 2018). However, these statistics are not the only indicators of travel fatigue. In the current study, we expand this examination by investigating the effect of travel on game outcome, points scored and allowed, completion, field-goal, and extra-point percentages, as well as interceptions, sacks, and punts. Based on past literature, we hypothesized that West Coast teams would gain a circadian advantage when traveling eastward, and when playing at times closer to the circadian peak performance time (i.e., during the afternoon).

Method

Data for individual games and performance statistics were acquired from Pro-Football-Reference, a publicly-available, sports database. Elo values were obtained from FiveThirtyEight, a website that focuses on statistical analysis of not only sports topics, but also economics, politics, and other fields. Specifically, data from the 1990-2019 seasons were collected for all five West Coast teams in the NFL, those being: the Los Angeles Rams, Los Angeles (formerly San Diego) Chargers, Oakland (currently Las Vegas) Raiders, San Francisco 49ers, and Seattle Seahawks. The NFL implemented a bye week in 1990, allotting teams a week with no games and thus no travel, so data were gathered prior to this addition. Furthermore, data from the 2020 season were not considered due to numerous players opting out of games as a result of the COVID-19 pandemic (Around the NFL Staff 2020). In addition, due to small group sample sizes, games were removed from our analysis if they were scheduled in the mornings, occurred on a Friday, or ended in a tie. We also removed games played internationally owing to the uniqueness of international travel and the competitive environment of such neutral site games. Of the 2,064 games initially collected, a total of 1,018 regular season away games were examined during this thirty-year period.

We investigated the impact of travel by examining the effects of time zone, local time of day, and their interaction on visiting team performance. Performance was measured by game outcome, points scored and allowed, completion, field-goal, and extra-point percentages, interceptions, sacks, and punts. Local game times were grouped into two categories: *afternoon* and *evening* (i.e., games with kick-off times before or after 5:00 p.m. local time, respectively). Time zones included the four United States time zones: Pacific, Mountain, Central, and Eastern. We also controlled for the day of the week when games were played, which included Mondays, Thursdays, Saturdays, and Sundays in our sample. To control for the strength of teams, we utilized the Elo value prior to each matchup for both the West Coast team and their opponent.

Data Analysis

All analyses were performed in RStudio (Version 1.4.1717; RStudio Team n.d.) and jamovi (Version 1.6.23; The jamovi project n.d.). We conducted a series of generalized regressions with Bonferroni-corrected post-hoc tests. A linear model was assumed for completion, field-goal, and extra-point percentages, while a Poisson model was implemented for points scored, points allowed, and punts. Due to over dispersion, a negative binomial model was utilized in the analysis of interceptions and sacks allowed. Finally, a logistic model was applied for game outcomes due to the binary nature of the measure (i.e., in our study, either a win or loss).

Results

Overall Results

A significant effect of time zone, time of game, and their interaction was detected for points scored (p values $< .01$). Time of game also meaningfully impacted points allowed by West

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Coast teams (p = .04), as well as completion percentage (p = .05). The logistic model for game outcome yielded a significant effect of time zone (p = .02). There was also compelling evidence for the effect of time zone on sacks allowed (p = .005). There were no meaningful differences across interceptions, punts, along with field-goal and extra-point percentages (p values > .05). A summary of these results is provided in Tables 1–3.

Table 1: Regression Results for Points Scored, Points Allowed and Game Outcome

	PS (R^2 = .08)		*PA* (R^2 = .11)		*GO* (R^2 = .09)	
Variable	χ^2	*p*	χ^2	*p*	χ^2	*p*
Time Zone	65.84	***	5.10	.16	10.19	.02
Time of Game	7.96	.005	4.04	.04	0.07	.79
Time Zone × Time of Game	44.05	***	2.92	.40	7.43	.06
Covariates						
Day of Game	5.25	.15	30.64	***	0.91	.82
Team Elo	255.20	***	148.01	***	56.16	***
Opponent Elo	112.98	***	284.83	***	61.87	***

Note. ***p < .001. *PS* = points scored. *PA* = points allowed. *GO* = game outcome

Table 2: Regression Results for Completion, Extra-Point, and Field-Goal Percentages

	Comp % (R^2 = .07)		*XP %* (R^2 = .01)		*FG%* (R^2 = .01)	
Variable	*F*	*p*	*F*	*p*	*F*	*p*
Time Zone	0.99	.40	0.34	.80	1.26	.29
Time of Game	4.02	.05	2.57	.11	1.76	.19
Time Zone × Time of Game	0.62	.60	0.63	.60	0.91	.43
Covariates						
Day of Game	2.32	.07	2.28	.08	1.26	.29
Team Elo	59.47	***	0.05	.83	0.03	.86
Opponent Elo	4.24	.04	0.37	.54	0.32	.57

Note. ***p < .001. *Comp %* = completion percentage. *XP %* = extra-point percentage. *FG %* = field-goal percentage

Table 3: Regression Results for Interceptions, Sacks Allowed, and Punts

	Ints (R^2 = .02)		*SA* (R^2 = .02)		*Pnts* (R^2 = .05)	
Variable	χ^2	*p*	χ^2	*p*	χ^2	*p*
Time Zone	1.52	.68	12.4	.006	7.08	.07
Time of Game	0.21	.64	1.52	.22	0.35	.56
Time Zone × Time of Game	0.35	.95	8.85	.03	5.47	.14
Covariates						
Day of Game	5.54	.14	6.62	.09	2.92	.40
Team Elo	4.43	.04	1.31	.25	28.32	***
Opponent Elo	11.01	***	3.10	.08	1.63	.20

Note. ***p < .001. *Ints* = interceptions. *SA* = sacks allowed. *Pnts* = punts

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Main Effects

Specifically, we found that West Coast teams scored more points in the Eastern time zone (Estimated marginal mean [*EMM*] = 21.65, standard error [*SE*] = 0.50) compared to the Pacific (*EMM* = 19.09, *SE* = 0.44; $z = 4.90$, $p < .001$) and Central time zones (*EMM* = 17.90, *SE* = 0.44; $z = 7.27$, $p < .001$). West Coast teams playing away games in the Mountain time zone (*EMM* = 21.19, *SE* = 0.56) also scored more points than those playing in the Pacific ($z = 3.50$, $p = .003$) and Central time zones ($z = 5.59$, $p < .001$). Teams also scored more points ($z = 2.80$, $p = .005$) during games played in the afternoon (*EMM* = 20.73, *SE* = 0.47) than the evening (*EMM* = 19.10, *SE* = 0.43).

Teams had a higher winning percentage ($z = 2.82$, $p = .03$) in the Eastern time zone (*EMM* = 0.50, *SE* = 0.06) than the Central time zone (*EMM* = 0.32, *SE* = 0.05). West Coast teams also allowed fewer points ($z = -2.02$, $p = .04$) during afternoon games (*EMM* = 21.49, *SE* = 0.45) compared to those played in the evening (*EMM* = 22.69, *SE* = 0.47). Teams also had higher completion percentages ($t = 2.00$, $p = .05$) during afternoon (*EMM* = 60.68%, *SE* = 1.06%) than evening games (*EMM* = 57.90%, *SE* = 1.06%). Ultimately, teams allowed more sacks ($z = 3.35$, $p = .005$) in the Central time zone (*EMM* = 2.64, *SE* = 0.19) than the Eastern time zone (*EMM* = 1.98, *SE* = 0.17).

Interaction Effect on Points Scored

Within the interaction between time zone and time of game, the results indicated that West Coast teams playing away evening games in the Eastern time zone and afternoon games in the Mountain time zone scored more points than teams playing during the afternoon and evening in the Central time zone. Teams playing in the Central and Mountain time zones during the afternoon scored more points than those playing during the evening in the Central and Pacific time zones. West Coast teams playing afternoon games in the Eastern and Pacific time zones scored more points than those playing in the Central time zone during the evening. Teams playing afternoon and evening games in the Eastern time zone, afternoon games in the Pacific time zone, and evening games in the Mountain time zone also scored more points than teams playing in the Pacific time zone during the evening. The full results of these comparisons are listed in Tables 4 and 5.

Table 4: Estimated Marginal Means for the Effect of Time Zone and Time of Game on Points Scored

				95% Confidence Interval	
Time of Game	***Time Zone***	***EMM***	***SE***	***Lower***	***Upper***
Evening	Eastern	22.80	0.81	21.27	24.44
Afternoon	Mountain	21.75	0.66	20.49	23.09
Afternoon	Pacific	20.93	0.57	19.84	22.08
Evening	Mountain	20.64	0.89	18.97	22.46
Afternoon	Eastern	20.56	0.50	19.60	21.57
Afternoon	Central	19.75	0.49	18.81	20.74
Evening	Pacific	17.42	0.60	16.28	18.63
Evening	Central	16.22	0.65	15.00	17.55

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Table 5: Post-Hoc Comparisons for the Effect of Time Zone and Time of Game on Points Scored

Time Zone	*Time of Game*		*Time Zone*	*Time of Game*	*Odds Ratio*	*SE*	*z*	*p*
Central	Evening	–	Eastern	Evening	0.71	0.03	-6.96	***
Mountain	Afternoon	–	Central	Evening	1.34	0.07	5.98	***
Eastern	Evening	–	Pacific	Evening	1.31	0.06	5.77	***
Pacific	Afternoon	–	Central	Evening	1.29	0.06	5.41	***
Eastern	Afternoon	–	Central	Evening	1.27	0.06	5.23	***
Mountain	Afternoon	–	Pacific	Evening	1.25	0.06	5.04	***
Central	Evening	–	Mountain	Evening	0.79	0.04	-4.42	***
Pacific	Afternoon	–	Pacific	Evening	1.20	0.05	4.37	***
Central	Afternoon	–	Central	Evening	1.22	0.06	4.32	***
Eastern	Afternoon	–	Pacific	Evening	1.18	0.05	4.15	***
Central	Afternoon	–	Mountain	Afternoon	0.91	0.02	-3.70	.006
Central	Afternoon	–	Eastern	Evening	0.87	0.04	-3.55	.01
Mountain	Evening	–	Pacific	Evening	1.19	0.06	3.23	.03
Central	Afternoon	–	Pacific	Evening	1.13	0.05	3.12	.05
Central	Afternoon	–	Pacific	Afternoon	0.94	0.02	-2.58	.27
Eastern	Afternoon	–	Eastern	Evening	0.90	0.04	-2.57	.28
Eastern	Afternoon	–	Mountain	Afternoon	0.95	0.02	-2.25	.69
Central	Afternoon	–	Eastern	Afternoon	0.96	0.02	-2.17	.84
Pacific	Afternoon	–	Eastern	Evening	0.92	0.04	-2.03	1.00
Eastern	Evening	–	Mountain	Evening	1.10	0.06	1.92	1.00
Central	Evening	–	Pacific	Evening	0.93	0.05	-1.42	1.00
Mountain	Afternoon	–	Pacific	Afternoon	1.04	0.03	1.38	1.00
Mountain	Afternoon	–	Eastern	Evening	0.95	0.04	-1.06	1.00
Mountain	Afternoon	–	Mountain	Evening	1.05	0.06	1.01	1.00
Central	Afternoon	–	Mountain	Evening	0.96	0.05	-0.90	1.00
Eastern	Afternoon	–	Pacific	Afternoon	0.98	0.02	-0.83	1.00
Pacific	Afternoon	–	Mountain	Evening	1.01	0.05	0.28	1.00
Eastern	Afternoon	–	Mountain	Evening	1.00	0.05	-0.08	1.00

Note. ***$p < .001$

Discussion

Findings

Over the thirty-year period examined, the data reflect that West Coast teams experienced the highest winning percentages in the Eastern time zone followed by the Mountain, Pacific, and Central time zones, respectively. Consistent with previous research (Smith, Guilleminault, and Efron 1997; Roy and Forest 2018), our findings generally support the trend that West Coast teams experience a circadian advantage when traveling eastward. Although West Coast teams experienced the highest win percentage in Eastern time zones, it is interesting to note that there

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is a visible drop-off in win percentage, as well as points scored in the Central time zone (i.e., the lowest of all four time zones). One potential explanation resides in the performance of teams in the Central time zone during this span. For instance, many of these teams have had consistently strong defensive units over the thirty-year period. Teams, such as the Kansas City Chiefs, Green Bay Packers, Minnesota Vikings, Chicago Bears, Dallas Cowboys, and New Orleans Saints, have often placed in the top half of rankings for team defense over the thirty-year period (Pro-Football-Reference n.d.). Although we did attempt to mitigate these issues by utilizing Elo ratings in our models, specific defensive metrics were not considered as covariates. In addition, there are only two NFL teams in the Mountain time zone, those being: the Arizona Cardinals and Denver Broncos. The Cardinals have consistently ranked in the bottom-half of the league, while the Broncos have fluctuated between the top- and bottom-half over the thirty-year period.

Our data also display an interaction between the time of game and time zone on points scored. Specifically, West Coast teams scored the most points during evening games played in the Eastern time zone. This finding supports our hypothesis because West Coast teams traveling eastward and playing in the evening are playing closer to their peak time for athletic performance (Facer-Childs, Boiling, and Balanos 2018). Overall, West Coast teams also scored more and allowed fewer points during games played in the afternoon compared to games played in the evening. In addition to points scored, points allowed, and win percentage, there was also a significant association between time zone and sacks allowed. West Coast teams allowed more sacks in the Central time zone than the Eastern time zone. Sacks are indicative of offensive performance as they refer to the number of failures the offensive line had in protecting the quarterback from being tackled. In the Eastern time zone, teams play closer to their peak performance time for strength, flexibility, and reaction time for afternoon and evening games (Smith, Guilleminault, and Efron 1997). Reaction time and strength both play major factors in allowing sacks, as offensive linemen must react quickly and utilize their strength to prevent defensive players from tackling or "sacking" the quarterback. Ultimately, they made fewer errors in protecting the quarterback in the Eastern time zone compared to that of the Central time zone, consistent with our hypothesis of the West Coast circadian advantage with increased eastward travel and previous research (Smith, Guilleminault, and Efron 1997; Roy and Forest 2018).

Our data did not reveal any meaningful impact of circadian disruption and travel on interceptions, punts, extra-point percentage, and field-goal percentage. This lack of association could be due to a variety of factors. For example, interceptions can be affected by footballs being dropped or tipped by receivers, the quarterback missing their target, quarterbacks being hit while throwing, or even a byproduct of certain offenses who are pass-heavy and thus create more opportunities for interceptions and missed throws. Alternatively, there may have been an effect of travel on sacks allowed due to sacks being a product of individual matchups: either the offensive linemen block the defensive player from tackling the quarterback, or the defensive linemen get by the offensive linemen resulting in a sack. In addition, field-goal and extra-point percentages can be affected by wind and other weather conditions, which vary across different stadiums. In our study, we did not account for weather conditions. Other potential confounding variables that were not controlled for could include the impact of injuries at other positions on performance and even playing surfaces (e.g., artificial turf or natural grass).

Limitations and Future Directions

Considering this, we came across several other limitations during this study. Due to the NFL's nature, there is a large disparity of teams in each time zone. For instance, two teams are in the Mountain time zone and seventeen are in the Eastern time zone (NFL 2021). As a result, it is difficult to accurately measure the effect of traveling through different time zones on performance because teams spend different amounts of time in each time zone. Additionally, we did not account for the changes in league rules and general play style over time. Some of these

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rule changes include modification of the distance for extra points, which decrease the likelihood and percentage of these field goals, elimination of chop blocks (NFL Operations 2021) to reduce player injury risk, and rules protecting the quarterback from hits to the head or players diving at the quarterback's knees (Weiner 2018), limiting concussion and general injury risk, all of which could affect performance. In addition, overall scoring has gradually increased over time, given that the 2020 season witnessed the most total points scored in all NFL games compared to any other season (Smith 2021).

Because of the limited sample of games played in the Mountain time zone, future research could examine teams that only play in the Mountain time zone, specifically in Denver. Because of Denver's elevated location, performance may be hindered for many players. It could be interesting to examine if this altitude exacerbates any disadvantages due to travel. Our findings also revealed a circadian advantage for teams traveling eastward, so it would also be interesting to perform a case study of solely East Coast teams to examine the deleterious effects of westward travel on performance. Further research could also observe the effect of travel and circadian disruption on injury risk on a weekly basis. Finally, it may also be useful to investigate the effect of international travel and circadian disruption on performance in the Olympics. International travel creates a much greater disparity in time difference compared to one's home base, which could therefore have more potent effects on performance than domestic travel within the NFL. However, it may be interesting to compare and contrast the effects of circadian disruption in different sports contexts to tease out variations in performance.

Concluding Points and Implications

Ultimately, West Coast teams in the NFL experienced a circadian advantage of heightened performance with eastward travel, scoring the most points and winning the most games in the Eastern time zone. In addition, playing in the afternoon was most advantageous for West Coast teams' offensive and defensive performance measured by scoring more points, allowing fewer points, and having higher completion percentages. Our findings support our hypothesis that West Coast teams would gain circadian advantages with eastward travel and playing at afternoon times, because athletes would play closest to their peak performance time of the late afternoon (Smith, Guilleminault, and Efron 1997; Facer-Childs, Boiling, and Balanos 2018).

West Coast teams' circadian advantages when traveling eastward and playing in the afternoon have significant implications on the current scheduling of games. East Coast teams playing at home most commonly play in the earliest time slot on Sundays at 1:00 p.m. Eastern Standard Time (EST). Thus, visiting West Coast teams may feel as if they are playing at 10:00am Pacific Standard Time (PST), further from the advantageous time of the late afternoon. Moreover, in an analysis of 374 NFL games from 2001 to 2015, a road team from the West or Mountain time zone playing in the East or Central time zones lost 64.40 percent of the games, exceeding the 57.20-percent win rate for home teams (Stonelake 2016). Thus, current scheduling patterns consign West Coast teams to play at a time unfavorable to their peak performance time. The NFL should consider addressing such disparities by scheduling games closer to the afternoon peak time for teams from different time zones. For example, the NFL could explore scheduling more games with teams from opposing time zones at 4:00 p.m. EST, when West Coast teams feel as if they are playing at 1:00 p.m. PST.

Furthermore, the NFL must be mindful of scheduling games at late time periods (evening games) or with limited rest periods (Thursday games), which not only hinder performance but also put the individual athlete's health at risk. Certain teams and players have implemented strategies to combat circadian disruption in the NFL, and the rest of the league should follow their examples. In the past, the New York Jets have made efforts to preset their temperatures in the hotel room to a sleep therapy recommended 68°F, or even resorted to sleeping in pitch black rooms (Clark 2012). Under coach John Harbaugh, the Baltimore Ravens implemented measures

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to combat jet lag by starting their practices and meetings later to better acclimate to West Coast time in Baltimore before traveling to the West Coast (Clark 2012). John's brother, former 49ers coach Jim Harbaugh, applied similar measures by having the team travel early prior to games to adjust to the new time zone (Clark 2012). However, many players do not follow elaborate strategies and simply go to bed early. For example, Quarterback Tom Brady has admitted that he would go to bed at 7:15 p.m. on Monday nights (Clark 2012). Technology has also revolutionized sleep hygiene, as many NFL teams have turned to wearables to serve as a sleep monitoring system (Tanier 2016).

The effect of circadian disruption on performance can also have implications on nutrition. For example, social jetlag, or the asynchrony between one's biological clock and social obligations or activities (Tavernier, Munroe, and Willoughby 2015), can contribute to the discrepancy in one's sleep pattern and circadian misalignment. In a study of 792 individuals, participants with social jetlag reported a higher intake of protein, total fat, total calories (kcal), saturated fat, cholesterol, servings of meat and eggs, and sweets compared to those without social jetlag (Mota et al. 2019). Participants experiencing social jetlag also reported later meal timing for breakfast and early afternoon snack and dinner (Mota et al. 2019). Circadian disruption as a result of traveling through different time zones and its resulting social jetlag has been associated with a weak diet and later-occurring mealtimes, serving as a risk factor for obesity-related diseases. These behavioral patterns could also be evident within athletes with lifestyles involving frequent travel. However, by altering one's meal timing and nutrition, athletes can work to effectively combat the effects of circadian disruption. A study examining the influence of meal timing on the circadian rhythm during a thirteen-day period revealed that delaying mealtimes by five hours can change the phase relationship of circadian rhythms (Wehrens et al. 2017). Regulating meal timing can regulate internal rhythms as one of the authors of the study, Jonathan Johnston, disclosed, "a 5-hour delay in mealtimes causes a 5-hour delay in our internal blood sugar rhythms" (Cell Press 2017). It is also possible that consuming tart cherries can help reduce jet lag by inducing sleep because tart cherries contain melatonin, which helps regulate the sleep-wake cycle (Adigun 2021). Alternatively, eating watermelon could potentially help feelings of fatigue and improve focus due to its high-water content at approximately 92 percent. In addition, avoiding coffee can be advantageous when dealing with jet lag because excess coffee can contribute to dehydration and delay the increase in melatonin, which can disturb one's sleep-wake cycles. Thus, by actively altering nutrition plans and meal timing, athletes can attempt to combat the detrimental effects of circadian disruption on their performance.

In light of the circadian effects of time zone and time of game, we recommend teams consciously embrace circadian advantages when presented with eastward travel. When faced with circadian disadvantages, we recommend athletes partake in methods previously utilized by NFL teams through circadian assimilation, such as travelling earlier to adjust to new time zones, sleeping early, utilizing sleep monitoring systems and other wearable technologies, sleeping in rooms with temperatures most suitable for sleep, and adjusting their diet accordingly. Overall, we hope our study inspires further research to investigate the relationship of sports performance with the effectiveness of these methods and other countermeasures in coping with circadian disruption.

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